

A Task-First Audit of Dynamic LiDAR World Simulation: Supplementary Material

Anonymous Authors

1. Data and Evaluation Protocol

The clean split is selected from dependency-chain-unexposed nuScenes logs using metadata only: four development logs, three route-selection logs, and three unopened source-test candidates. Selective archive I/O materializes 3,882 development and route-selection keyframe scans (2,695,516,800 bytes). Only development is compiled and evaluated after the gate is frozen. It contains 65 scenes, 501 Actors, 2,059,365 build points, and 641,930 held-out positive-return rays. Actor bundles preserve point/ray provenance, intensity, ring, time, and poses; target arrays are stored separately.

2. Formal Runs

The AdaPoinTr transfer baseline uses the official PCN checkpoint, followed by 600 epochs of legacy adaptation. A1 starts from that frozen checkpoint and runs 100 additional epochs with surface, hit, and known-free losses. Its checkpoint SHA-256 is d489e2bb . . . e2d7; the full digest is retained in the run manifest. The clean development evaluator compares raw fusion, Actor TSDF, adapted AdaPoinTr, unanchored A1, and the fixed anchored output at 64, 128, 256, 512, and native budgets. Every run records a resolved configuration, manifest, status, Actor and log rows, aggregate summary, resource use, data roles, and failure-ledger references.

3. D1 Gate

At the 512-point cap, Actor TSDF and the anchored candidate obtain CD .178599/.194976, F-score .750319/.714542, early rate .411462/.439344, and hit recall .549722/.522150. The candidate therefore misses both primary-effect alternatives and both side-effect limits. The mechanical result is `decision=close_method.claim`. Route selection, source test, and external test remain unread, and no P3/P4 experiment is authorized.

4. Failure and Scope Boundary

V71-F62 records the recoverable clean-data payload assumption; selective official-archive extraction resolves it

without changing the split. V71-F63 records the scientific D1 rejection. The latter is not repaired with an anchor-fraction, loss-weight, density, or threshold sweep. LiDAR4D demonstrates full-scan capability on its native KITTI-360 protocol, but its absolute numbers are not compared with the Actor task.

5. Reproducibility Checklist

The final role state is: legacy diagnostic and clean development read; route selection, source test, and external test unread. Data, method, and metric artifacts are linked in `docs/WORLDSIM_V7_2_D1_NEGATIVE_CLOSEOUT.md`. The main paper and this supplement describe a negative technical report rather than a submission-ready method.